ABSTRACT

Our current supply chains struggle with hidden inefficiencies and vulnerabilities due to limited transparency. This opacity fuels the rampant issue of counterfeiting, often undetectable by sight. Existing methods like RFID tags and AI offer some solutions, but they're hampered by limitations like replicability and high computational demands. This project proposes a novel approach: harnessing the power of blockchain technology. By creating a decentralized, tamper-proof ledger that tracks every step of a product's journey, this system guarantees transparency and traceability throughout the supply chain. Blockchain based system, makes everything decentralized that may be accessed by several parties at the same time. One of its main advantages is that the recorded data is difficult to change without the consent of all parties concerned which makes the data extremely secure and protect from all vulnerabilities. This project presents system designed using blockchain technology for detection of counterfeit products.